POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

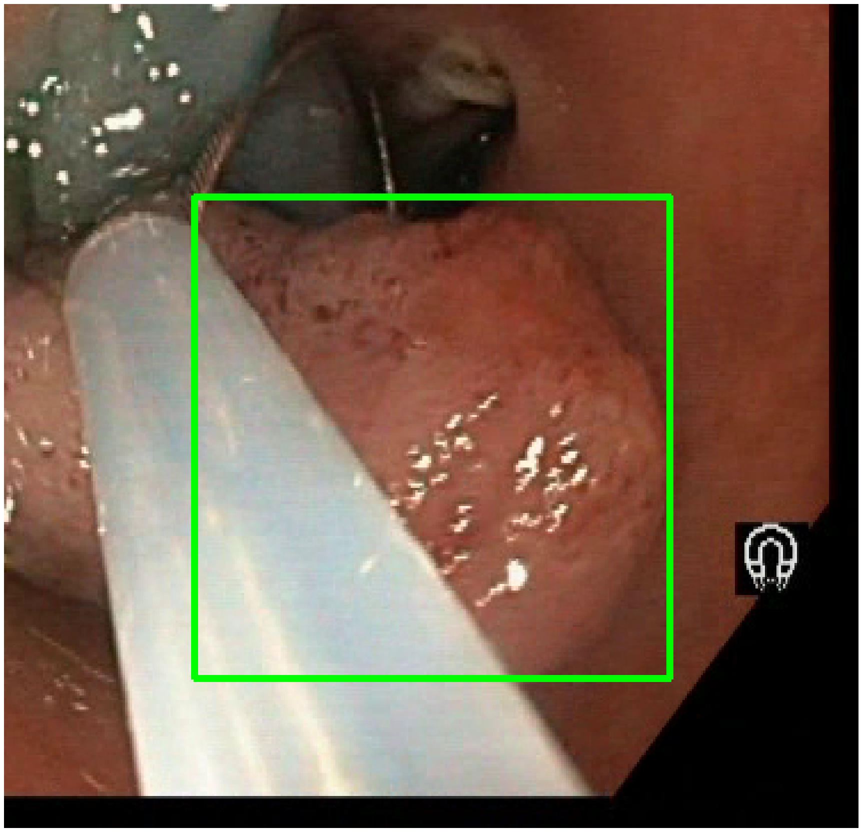
Video ID:	a3e1c30d-4522-4170-b056-a937a6d76748
Total Frames:	1450
Frame Rate:	25.0 fps (assumed)
Duration:	58.0 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	1059	✓
RT-DETR Detections	1220	✓
MedSAM2 Detections	1211	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	0	0.0%
MEDIUM RISK	3	100.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 866 frames)



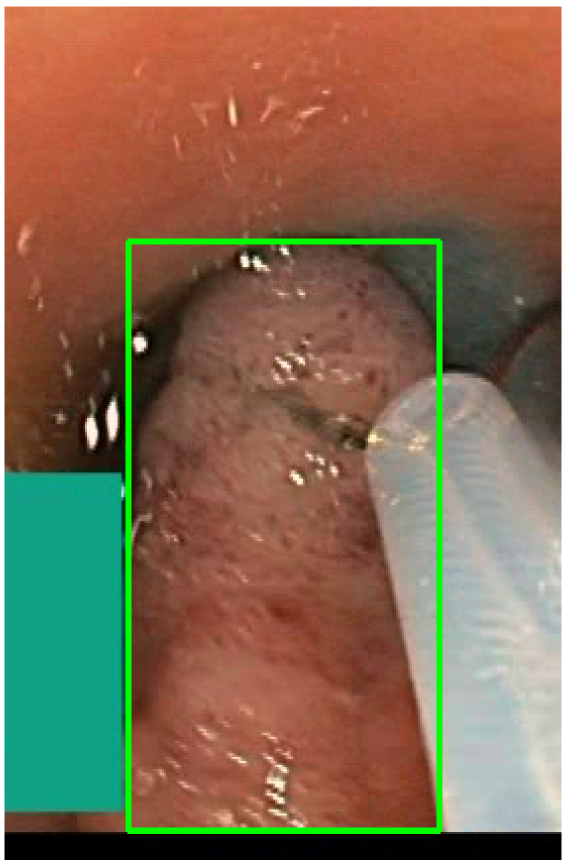
Feature	Value	Interpretation
Frame Index	463	Frame 463 of 1450
Detection Method	Detector Consensus (YOLO + RT-DETR agreement) (MedSAM2 unavailable for this video)	MedSAM2 Unavailable
POLYP TYPE	**FLAT_POLYP**	Type confidence: 92.0%
Redness Score	0.151	Color-based vascularization
Relative Radius	26.3%	Polyp size as % of frame diagonal
Texture Score	0.301	Surface morphology
Vessel Visibility	0.430	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket

Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	93.8%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #2 — FLAT_POLYP [MEDIUM_RISK] (Tracked across 22 frames)



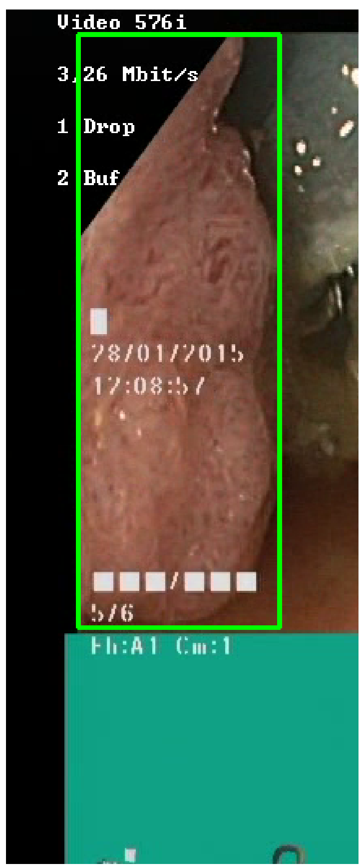
Feature	Value	Interpretation
Frame Index	336	Frame 336 of 1450
Detection Method	Detector Consensus (MedSAM2 Unavailable)	YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)
POLYP TYPE	**FLAT_POLYP**	Type confidence: 92.0%
Redness Score	0.125	Color-based vascularization

Relative Radius	15.0%	Polyp size as % of frame diagonal
Texture Score	0.244	Surface morphology
Vessel Visibility	0.599	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	LOW_RISK	Final symbolic reasoning bucket
Expert Prediction	LOW_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	88.7%	YOLO / RT-DETR / track confidence

Type Assessment: Flat mucosal lesion — chromoendoscopy or NBI assessment recommended.

Low-confidence MODERATE RISK detection - Clinical review recommended

POLYP #3 — SERRATED_POLYP [MEDIUM_RISK] (Tracked across 44 frames)



Feature	Value	Interpretation

Frame Index	1338	Frame 1338 of 1450
Detection Method	Detector Consensus (YOLO + RT-DETR agreement (MedSAM2 unavailable for this video)	(MedSAM2 Unavailable)
POLYP TYPE	**SERRATED_POLYP**	Type confidence: 92.0%
Redness Score	0.128	Color-based vascularization
Relative Radius	36.4%	Polyp size as % of frame diagonal
Texture Score	1.000	Surface morphology
Vessel Visibility	0.701	Vascularization indicator
Risk Tier	MEDIUM RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	81.4%	YOLO / RT-DETR / track confidence

Type Assessment: Serrated lesion — consider serrated polyposis syndrome. Complete resection required.

Low-confidence MODERATE RISK detection - Clinical review recommended

CLINICAL IMPRESSION

Summary:
A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 0 (0.0%)
• MEDIUM RISK polyps: 3 (100.0%)
• LOW RISK polyps: 0 (0.0%)

Recommendation:
The presence of MEDIUM RISK polyps warrants clinical review and correlation with endoscopic findings.
Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

Disclaimer: *This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.*

AI-Based Polyp Detection System
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